

## Analysis First-Year Exam, August 2016, Part 2

---

Answer exactly *FOUR* questions. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

1. Let the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  be uniformly continuous. For each positive integer  $n$  let  $f_n(t) = f(t + \frac{1}{n})$  whenever  $t$  is a real number. Prove that the sequence  $(f_n)_{n=1}^\infty$  is uniformly convergent. Give an example (with justification) to show that the conclusion can fail if the hypothesis uniformly is dropped.
2. Let  $A = \{a_n : n \geq 1\}$  be a countably infinite subset of  $[0, 1]$  and let  $1_A : [0, 1] \rightarrow \mathbb{R}$  be its indicator (or characteristic) function. Exhibit such a set  $A$  for which  $1_A$  is not Riemann integrable and exhibit such a set  $A$  for which  $1_A$  is Riemann integrable, providing justification in each case.
3. Let  $(f_n)_{n=1}^\infty$  be a sequence of continuous functions from  $[0, 1]$  to  $[0, 1]$ . Prove that if  $f_n(t)$  decreases to 0 whenever  $0 \leq t \leq 1$ , then  $\int_0^1 f_n(t) dt \rightarrow 0$ . Does the same conclusion follow when decreases is replaced by converges? Justify.
4. Prove that if the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is differentiable then its derivative is measurable.
5. Let  $f$  be a Lebesgue integrable function on  $\mathbb{R}$  and let  $g$  be defined on  $\mathbb{R}$  by  $g(y) = \int_{-\infty}^\infty \cos(xy)f(x) dx$ . Prove that  $\lim_{k \rightarrow \infty} g(k) = 0$ .